

The OLS estimators of the true values β_0 and β_1 in the linear regression model are obtained by solving the following minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2. \quad (1)$$

To solve the minimization problem, we first take the partial derivative with respect to b_0 :

$$\begin{aligned} \frac{\partial}{\partial b_0} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 &= \sum_{i=1}^n \frac{\partial}{\partial b_0} (Y_i - b_0 - b_1 X_i)^2 \\ &= \sum_{i=1}^n 2(Y_i - b_0 - b_1 X_i)(-1) \\ &= -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i). \end{aligned} \quad (2)$$

Then, we take the partial derivative with respect to b_1 :

$$\begin{aligned} \frac{\partial}{\partial b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 &= \sum_{i=1}^n \frac{\partial}{\partial b_1} (Y_i - b_0 - b_1 X_i)^2 \\ &= \sum_{i=1}^n 2(Y_i - b_0 - b_1 X_i)(-X_i) \\ &= -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) X_i. \end{aligned} \quad (3)$$

The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the values of b_0 and b_1 for which (2) and (3) equal zero:

$$-2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad (4)$$

$$-2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0. \quad (5)$$

which are equivalent to

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad (6)$$

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0. \quad (7)$$

For (6), we note that for the left-hand-side,

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = \frac{1}{n} \sum_{i=1}^n Y_i - \frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 - \frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 X_i = \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}, \quad (8)$$

where $\bar{Y}$ and $\bar{X}$ denote the sample average of Y and X , respectively. This implies that

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \quad (9)$$

Similarly, for (7), we note that for the left-hand-side,

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 X_i - \frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 X_i^2 = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2. \quad (10)$$

Together with (9), this implies that

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0, \quad (11)$$

which is equivalent to

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} = \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \right). \quad (12)$$

The conclusion follows.